Hena Kharwa

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PROFESSIONAL SUMMARY

Founder of Agentlane, a control plane for multi-agent governance, orchestration, and coordination, focused on safe, auditable, verifiable collaboration between autonomous coding agents. AI Engineer with 3+ years of experience building, fine-tuning (PEFT/LoRA), and deploying production LLMs, RAG systems, and agentic workflows using PyTorch, LangChain/LangGraph, MCP, and FastAPI. Skilled in shipping secure, low-latency AI infrastructure across AWS and Azure (Docker, Kubernetes, CI/CD).

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Swift, HTML/CSS, Shell Scripting
ML & Deep Learning: PyTorch, CUDA, TensorFlow, Keras, Scikit-learn, ONNX, Mixed-Precision Training, OpenCV, ResNet-18, Recommender Systems, A/B Testing
NLP & LLMs: BERT, DistilBERT, Word2Vec, Llama-3, OpenAI GPT-5, Claude, Hugging Face Transformers, LoRA/PEFT Fine-Tuning, RAG, Knowledge Graphs
Agentic AI: LangChain, LangGraph, LangSmith, Model Context Protocol (MCP), Multi-Agent Orchestration, Agent Governance, Claude Code, OpenCode, Tool Calling, Human-in-the-Loop Agents, GitHub/Slack/Linear/Notion/Figma APIs
Web & Mobile: Next.js, React, Node.js, FastAPI, Flask, Quart, WebSockets, REST APIs, Streamlit, SwiftUI, Stanford Spezi, AssemblyAI, AVSpeechSynthesizer
Security & DevOps: Docker, Kubernetes, Caddy, GitHub Actions, CI/CD, RBAC, OAuth 2.0/PKCE, AES-256-GCM, scrypt, Rate Limiting, HIPAA, FHIR
Cloud & Data: AWS (EC2, S3, Lambda, SageMaker, Bedrock), Azure (Azure ML, AI Foundry, Containers), GCP, PostgreSQL, Redis, Neo4j, Firebase, FAISS, Pinecone

WORK EXPERIENCE

Founder Aug 2026 - Present

AgentLane – Control Plane for Agent Orchestration, Governance & Coordination *New York, USA*

- Architected a vendor-neutral control plane for multi-agent orchestration, coordinating concurrent Claude Code and OpenCode sessions against a shared GitHub repository
- Designed a governance policy engine gating every AI agent action behind human-approved, auditable authorization, enforcing scoped permissions per repository and agent
- Engineered a cross-agent coordination workflow, "Change Impact Room," that surfaces producer-consumer API conflicts between agents pre-merge, before they reach CI
- Defined versioned compatibility contracts and an automated verification gate to resolve cross-agent conflicts, scoping a 2-5 developer pilot across two agents on a TS/JS repo

AI Engineer Apr 2025 - May 2026

Stevens Institute of Technology *New Jersey, USA*

- Raised compliance anomaly-detection precision to 96% from 81% BERT baseline across 10K regulatory documents by engineering retrieval-augmented pipelines
- Minimized hallucinations to 10% on domain-specific compliance queries by fine-tuning Llama-3 8B leveraging LoRA adapters with RAG-grounded generation
- Scaled processing to 500K+ documents monthly at sub-200ms p95 latency by architecting event-driven AWS Lambda/S3/EC2 high-throughput ingestion pipelines
- Improved retrieval precision 22% over dense-only search by integrating Knowledge Graph entity-relationship links into hybrid FAISS/Pinecone vector retrieval

AI Engineer Nov 2025 - May 2026

Enlighten Infosystems *Remote*

- Decreased multi-step LLM task latency 35% by re-architecting production GenAI workflows as modular, reusable LangChain/LangGraph agentic orchestration pipelines
- Improved task accuracy 28% on domain-specific evaluation benchmarks by fine-tuning LLaMA models with PEFT/LoRA on business-specific training corpora
- Raised retrieval precision 30% across enterprise knowledge repositories by engineering production RAG with Hugging Face embeddings and FAISS/Pinecone
- Reduced end-to-end inference latency 25% for production real-time LLM applications by deploying containerized FastAPI GenAI microservices on AWS-Lambda

AI/ML Engineer Dec 2023 - Jul 2024

Dhyey Technologies - TechXi *Gujarat, India*

- Elevated production-model accuracy from 58% to 81% while shortening training duration 30% through architectural retuning and mixed-precision TensorFlow
- Increased semantic document-retrieval MRR@10 by 25% over TF-IDF baselines by fine-tuning BERT embeddings on 50K domain-specific engineering documents
- Decreased inference latency 35% across 2K+ requests/day by migrating synchronous inference-serving pipelines to asynchronous FastAPI on Azure Containers
- Accelerated model deployments from 3-day to 5-hour cycles by automating end-to-end ONNX-based GitHub Actions/Azure ML CI/CD release-automation pipelines

AI/ML Engineer Sep 2022 - Nov 2023

AtliQ Technologies *Gujarat, India*

- Cut recommendation errors 25% across 200K+ catalog products by building a hybrid collaborative-filtering and content-based PyTorch/CUDA recommendation engine
- Increased click-through rate 20% and conversions 5% across 100K-MAU A/B tests by replacing rule-based recommendations with BERT/Word2Vec embeddings
- Improved top-5 recommendation accuracy 18% over keyword-based retrieval by indexing 500K+ DistilBERT product embeddings with high-performance FAISS search
- Sustained 50K+ inference requests/day at sub-150ms p99 latency by deploying containerized FastAPI recommendation microservices on Azure ML endpoints

AI/ML Intern May 2022 - Aug 2022

Enlighten Infosystems *Gujarat, India*

- Reduced decision-making time 31% and improved prediction accuracy 10% by developing a PyTorch/CUDA expert system for decision-classification workflows
- Improved visual inspection accuracy 15% and reduced per-image latency 800ms to 600ms by building ResNet-18/OpenCV real-time computer-vision pipelines
- Delivered real-time ML predictions to 15 cross-functional stakeholders with 90% satisfaction by building interactive Streamlit dashboards and visualizations
- Reduced summary generation time 20% and improved F1 15% across 5K regulatory-compliance documents by prototyping BERT-based auto-summarization pipelines

PROJECTS

Nexus - Shared AI Workspace for Human-Agent Teams | TypeScript, Next.js, PostgreSQL, WebSockets, MCP

- Built Nexus solo from idea to production by architecting a real-time, scalable multiplayer workspace on WebSockets integrating GitHub, Slack, Linear, Notion, and Figma
- Gated 100% of agent writes behind human-approval, validated by 583 tests, by implementing Admin/Editor RBAC, audit-logging, and AES-256-GCM encrypted tokens
- Differentiated from Copilot, Cursor, and Devin by shipping cross-agent governance, one-click audit-log revert, and exactly-once workflow execution via Postgres row-locks

Hospitalist Companion App | 1st Place - Rutgers Health Hack 2025 (RWJUH Spezi App Track)

- Won 1st-place and doubled physicians' bedside diagnostic query speed by architecting an AI-powered clinical iOS application using SwiftUI/Stanford-Spezi/GPT-5
- Boosted semantic retrieval context accuracy 35% by integrating secure GPT-5/AWS Bedrock RAG pipelines on FHIR-compliant, HIPAA-ready cloud infrastructure
- Enabled hands-free clinical voice queries for physicians by integrating AssemblyAI and AVSpeechSynthesizer, secured through S3/Firebase and RBAC authentication

AI-Driven Personalized Alumni Engagement System | Azure AI Foundry, LangChain, React, Docker

- Connected students with alumni across 50K+ structured profiles by building high-performance Python Quart/Node.js backends on FAISS vector-similarity search
- Delivered personalized multi-turn recommendations across 50K+ alumni profiles by orchestrating LLM-driven workflows with LangChain and Azure AI Foundry
- Supported real-time student engagement by shipping responsive React web applications on Azure serverless infrastructure with Docker/GitHub CI/CD pipelines

PrepRight: AI Powered Interview Prep App | Python, React, Flask, BERT, FAISS, HuggingFace

- Improved interview-answer evaluation accuracy 25% by fine-tuning BERT/GPT transformer models with Hugging Face on specialized domain-specific interview datasets
- Increased resume-to-job matching precision 25% across 1K+ resumes and job descriptions by vectorizing them with FAISS-indexed dense-semantic embeddings
- Automated interview assessments end-to-end by building resume parsers, feedback modules, performance-analytics dashboards with React.js/Flask REST APIs on AWS

EDUCATION

Stevens Institute of Technology

Masters of Science, Applied Artificial Intelligence, GPA: 4.0/4.0 *Sep 2024 - May 2026*

Charotar University of Science and Technology

Bachelors of Technology, Computer Engineering, GPA: 3.84/4.0 *Oct 2020 - May 2024*

PUBLICATIONS

Hybrid InceptionNet Based Enhanced Architecture for Isolated Sign Language Recognition

https://ieeexplore.ieee.org/abstract/document/10577129